

CPE 486/586: Deep Learning for Engineering Applications

07 Variational AutoEncoder

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Outline

1. Motivation
2. AutoEncoder

Motivation

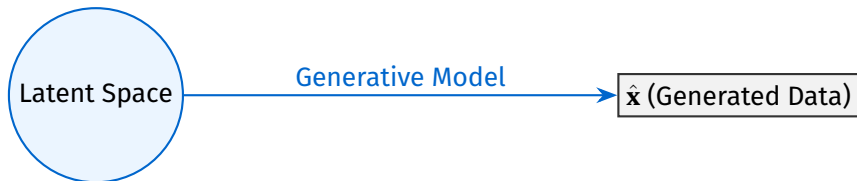
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0	1	1	1	1	1	1	1	0	1	0	1	0	1	1	0	1	1	0	1

Estimating Probability Distributions for Generative Processes

Given $\mathbf{x}$, we aim to estimate $P(\mathbf{x})$ to provide a complete description of the probabilistic model that generates the observed data.

- ⚡ **Data Synthesis:** We want to learn generative processes capable of producing new datasets.
- ⚡ **Noise Reduction:** By discarding noise, we can more effectively learn underlying physical laws.
- ⚡ **Causality:** Generative models provide a framework to express and understand causal relations.

Example: *If we understand the generative process of wildfires, we can apply that knowledge across different regions, such as California or Australia.*

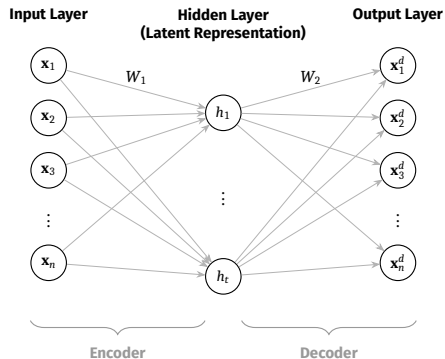


In this lecture, we model generative processes using Variational Autoencoders (VAEs).

AutoEncoder

0	0	1	0	0	1	1	0	1	0	1	1	1	1	1	1	1	1	0	0
0	0	1	0	0	0	1	1	0	0	0	0	1	1	1	0	0	1	1	0
0	1	1	0	1	1	1	1	1	0	1	0	0	1	1	0	1	0	1	0

What is AutoEncoder



Loss function is generally reconstruction loss:

$$\mathcal{L} = \frac{1}{n} \sum_{i=1}^n ||\mathbf{x}_i - \mathbf{x}_i^d||_2^2$$

An **autoencoder** is a neural network, whose main job is to encode the input into a compressed representation, and then decode it back such that the reconstructed input is similar as possible to the original one.

It has three parts:

- 1 **Encoder:** Learns the most relevant features and transfers data into a latent space.
- 2 **Latent representation:** a middle layer that represents feature in some low dimensional space.
- 3 **Decoder:** it takes the reduced features from latent space and lift to the original dimension.

As it turns out, AutoEncoder is a neural network with a bottleneck.

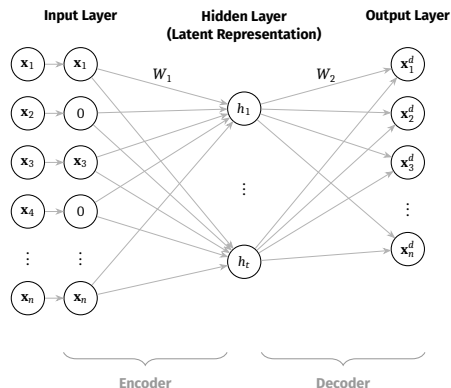
Denoising Autoencoder

An AutoEncoder can end up learning an identity function (also called null function) where input=output.

To remedy, we add some noise to the input to corrupt it before feeding to AutoEncoder. Such an AE is called denoising AutoEncoder (DAE).

There are two ways to add noise to DAE:

- 1 Add Gaussian noise to the input.
- 2 Randomly set some input nodes to zero.
- 3 Masking out a certain percentage of the input's pixels in the case of image data.



Stacked AutoEncoder

Reading List

- 1 Vincent, P., Larochelle, H., Bengio, Y., & Manzagol, P. A. (2008, July). Extracting and composing robust features with denoising autoencoders. In Proceedings of the 25th international conference on Machine learning (pp. 1096-1103).
<https://dl.acm.org/doi/pdf/10.1145/1390156.1390294>
- 2 Alain, Guillaume, and Yoshua Bengio. "What regularized auto-encoders learn from the data-generating distribution." The Journal of Machine Learning Research 15, no. 1 (2014): 3563-3593.
<https://www.jmlr.org/papers/volume15/alain14a/alain14a.pdf>